
TCX2005: Information Systems, Management and Organisations

L10: Building Information Systems

Prakash C. Sukhwai (Dr.)
Senior Lecturer, DISA
pcsdisa@nus.edu.sg

Lesson Objectives



How does building new systems produce organizational change?



Core activities & principal methodologies for systems development



GenAI and new era of product development

Learning Resources



- Textbook (read before class)
 - Read chapter 13
- Any Other Material
 - Shared on Canvas for this week

Case Study

Read the case study from textbook on –Vinci Energies Builds a New Mobile Employee Time Sheet App:

Identify:

- a. The problem
- b. Key Reasons or Causes
- c. Solution and its impact
- d. Learnings

Note: be ready to share your thoughts

Systems Development and Organizational Change

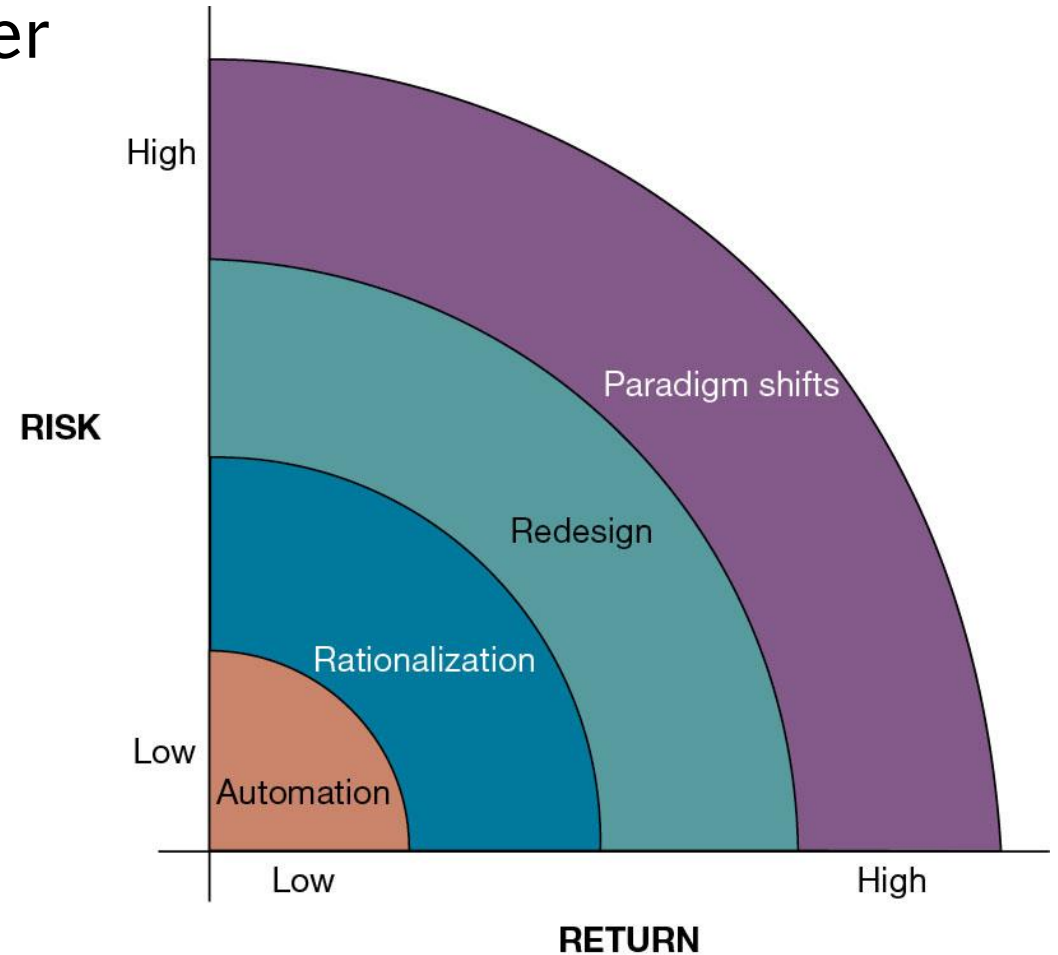
- **Automation:** Increases efficiency, replaces manual tasks
 - Example: The MOBITIME app automated the manual time sheet and project cost tracking process, replacing the need for field employees to physically return to the office to enter data into the SAP system (in the case study)
- **Rationalization of Procedures: Streamlines standard operating procedures**
 - Example: The new app streamlined the existing data entry process by allowing employees to input all time sheet data and project costs using a single application that directly transmits information to the SAP S/4HANA system (in the case study)

Systems Development and Organizational Change (cont.)

- **Business Process Redesign:** Analyzes, simplifies, and redesigns business processes
 - Example: The entire workflow for project tracking was redesigned—instead of traveling back to the office to enter data, field employees could now capture and submit information from any location, fundamentally changing how the company managed its 400,000 worldwide projects (in the case study)
- **Paradigm shifts:** Rethinks nature of business, defines new business model
 - Example: While not a complete paradigm shift, Vinci took significant steps toward mobile-first operations by moving from office-centric data entry to field-based reporting. This fundamentally changed how the company managed project tracking across 55 countries. (in the case study)

Organizational Change Carries Risks and Rewards

- Higher levels of change bring greater rewards but higher risks
- Most organizations aim for modest changes with acceptable risk levels



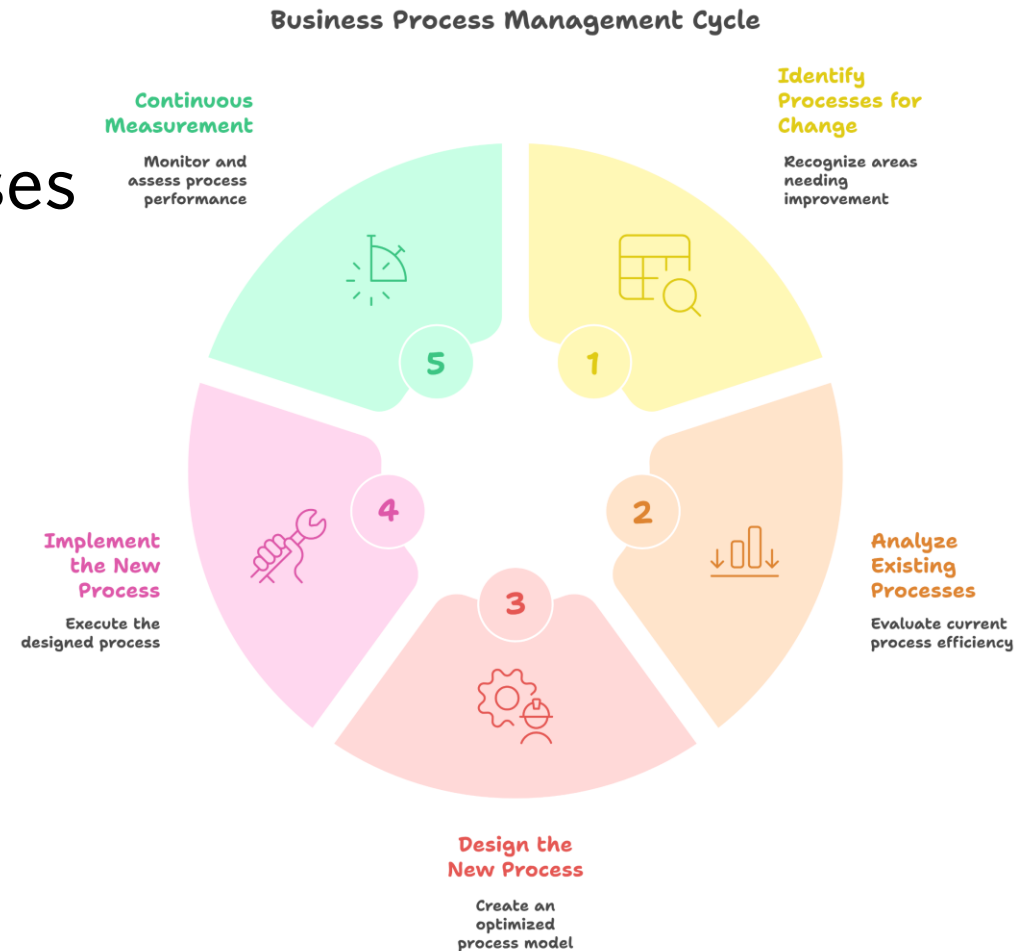
Business Process Management (BPM)

Business process management (BPM)

- Tools and methodologies to analyze, design, and optimize business processes
- Used by firms to manage business process redesign

Steps in BPM

1. Identify processes for change
2. Analyze existing processes
3. Design the new process
4. Implement the new process
5. Continuous measurement



Business Process Redesign

Example: Book Purchasing

As-Is Process (Physical Bookstore)

1. Customer travels to bookstore
2. Browses shelves for desired book
3. Takes book to checkout counter
4. Cashier processes payment
5. Customer returns home with book

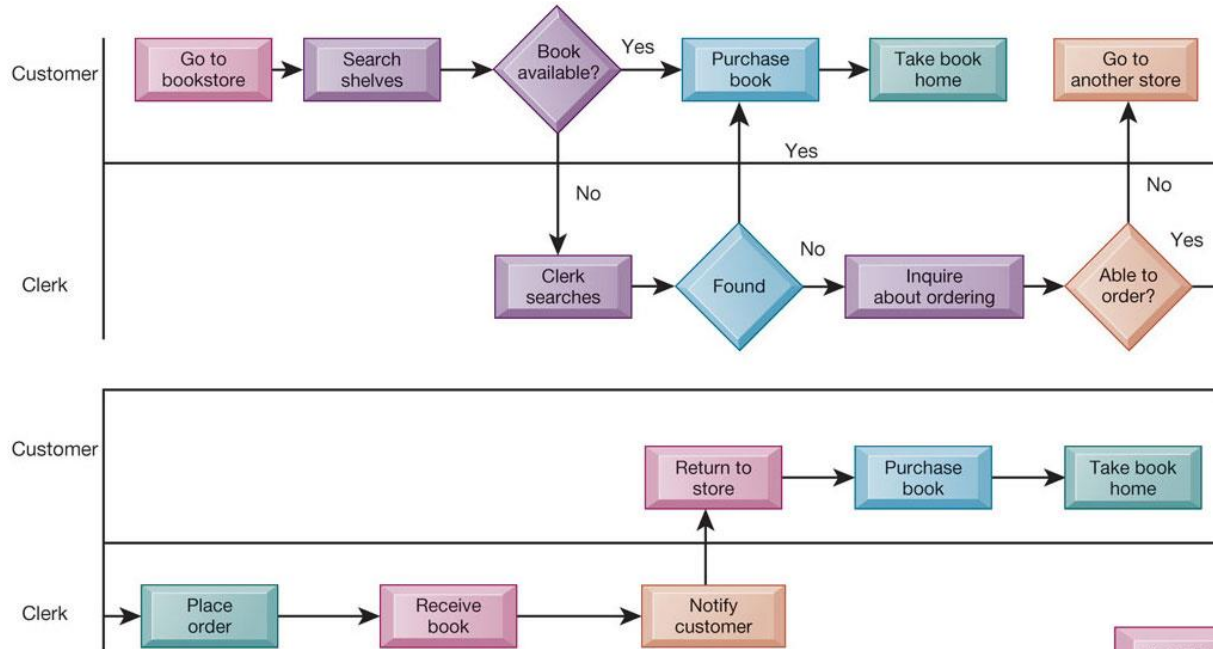
Redesigned/To-Be Process (Online)

1. Customer visits website
2. Searches for book
3. Places order online
4. System processes payment
5. Book is delivered to customer

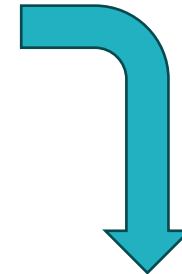
Another example: DP World's Process Redesign (textbook case-study)

Business Process Redesign (cont.)

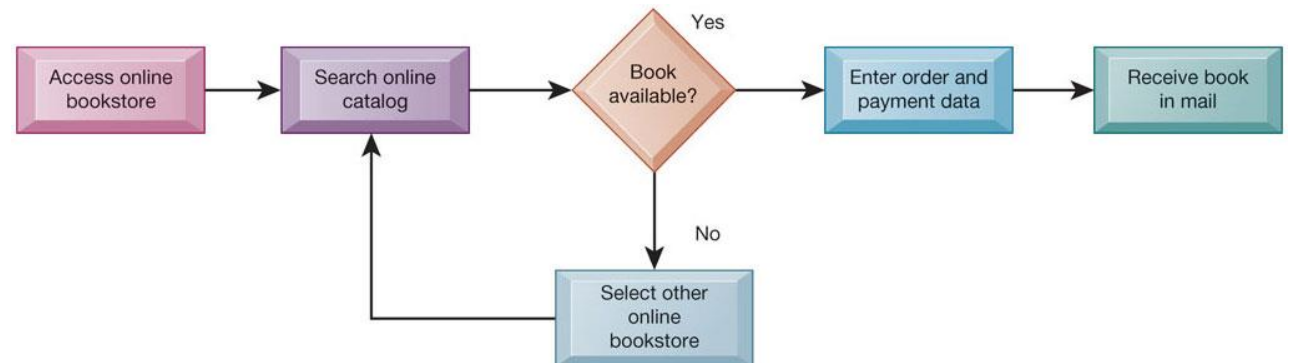
Book Purchasing



As-Is Process

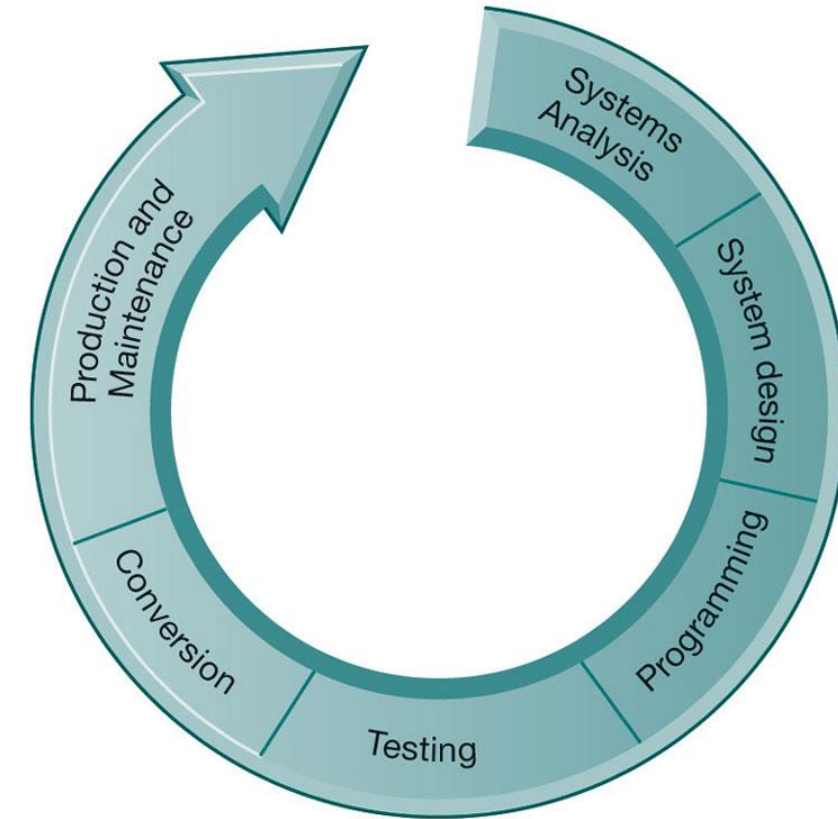


Redesigned/To-Be Process



The Systems Development Process

1. **Systems Analysis:** Define the problem, identify causes, specify solutions
2. **Systems Design:** Create detailed specifications
3. **Programming:** Translate designs into code
4. **Testing:** Ensure system produces correct results
5. **Conversion:** Switch from old system to new system
6. **Production & Maintenance:** Operate, evaluate, and modify the system



Systems Analysis

- Analysis of problem to be solved by new system
 - Defining the problem
 - Identifying causes
 - Specifying solutions
 - Identifying information requirements
- **Feasibility study**: Determines if solution is achievable
- **Systems proposal report**: Documents analysis findings
- **Information requirements**: Critical for successful implementation

Systems Design

- Describes system specifications that will deliver functions identified during systems analysis
- Addresses managerial, organizational, and technological components of system solution
- Role of end users
 - User information requirements drive system building
 - Users must have sufficient control over design process to ensure system reflects their business priorities and information needs
 - Insufficient user involvement in design effort is a major cause of system failure

Systems Design Specifications

Category	Specifications	Category	Specifications
Output	Medium, Content, Timing	Security	Access controls, Catastrophe plans, Audit trails
Input	Origins, Flow, Data entry	Documentation	Operations documentation, Systems documents, User documentation
User Interface	Simplicity, Efficiency, Logic, Feedback, Error handling	Conversion	Transfer files, Initiate new procedures, Select testing method; Cut over to new system
Database Design	Logical data model, Volume requirements, File organization	Training	Select training techniques, Develop training modules, Identify training facilities
Processing	Computations, Program modules, Required reports	Organizational Changes	Task redesign, Job redesign, Process design, Organization structure design, Reporting relationships
Controls	Input controls, Processing controls, Output controls		
Manual Procedures	What activities, Who performs them, When, How, Where		

Programming and Testing

- **Programming:** System specifications from design stage are translated into software program code
- **Testing**
 - Ensures system produces right results
 - Unit testing: Tests each program in system separately
 - System testing: Test functioning of system as a whole
 - Acceptance testing: Makes sure system is ready to be used in production setting
 - Test plan: All preparations for series of tests

System Conversion Strategies

- Process of changing from old system to new system
- Four main strategies
 - Parallel strategy: Run old and new systems simultaneously
 - Safest but expensive
 - Direct cutover: Switch from old to new system immediately
 - Risky but fast
 - Pilot study: Introduce system to limited area first
 - Reduces organizational risk
 - Phased approach: Introduce system in stages
 - By functions or organizational units
- Requires end-user training
- Finalization of detailed documentation showing how system works from technical and end-user standpoint

Case Study: NUHS System Transition

Project Overview



- National University Health System (NUHS) partnered with Integrated Health Information Systems (iHiS)
- Implemented a "mobile first" strategy
- Developed OneNUHS mobile app as their "digital front door"
- 12 months to conceptualize and develop the first two versions

Case Study: NUHS System Transition (cont.)

System Conversion Strategy

- Deployment approach:
 - Pilot study strategy to test with limited user groups
 - Phased approach strategy for broader implementation
- Development method:
 - Minimum Viable Product (MVP) principle
 - MVP 2: Added bill payment and caregiver access features
 - MVP 3: Added teleconsultation, COVID-19 records viewing, etc.
- Went live in December 2020

*One*NUHS
APP
Healthcare Made Simple



Case Study: NUHS System Transition (cont.)

Key System Conversion Principles Demonstrated

- **Pilot study:** Limited introduction before full deployment
- **Phased approach:** Gradual introduction of functionality
- **Iterative development:** Sequential MVP releases with expanding features
- **Digital transformation:** Mobile-first approach to healthcare services



Production and Maintenance

- System is reviewed for revisions
- May include post-implementation audit document
- **Maintenance activities:** Changes in hardware, software, documentation, or procedures to a production system to correct errors, meet new requirements, or improve processing efficiency
 - 20% debugging, emergency fixes
 - 20% changes to hardware, software, data reporting
 - 60% user enhancements, improving documentation, optimization
- **Post-implementation audit:** Evaluates system success

System Development Methodologies

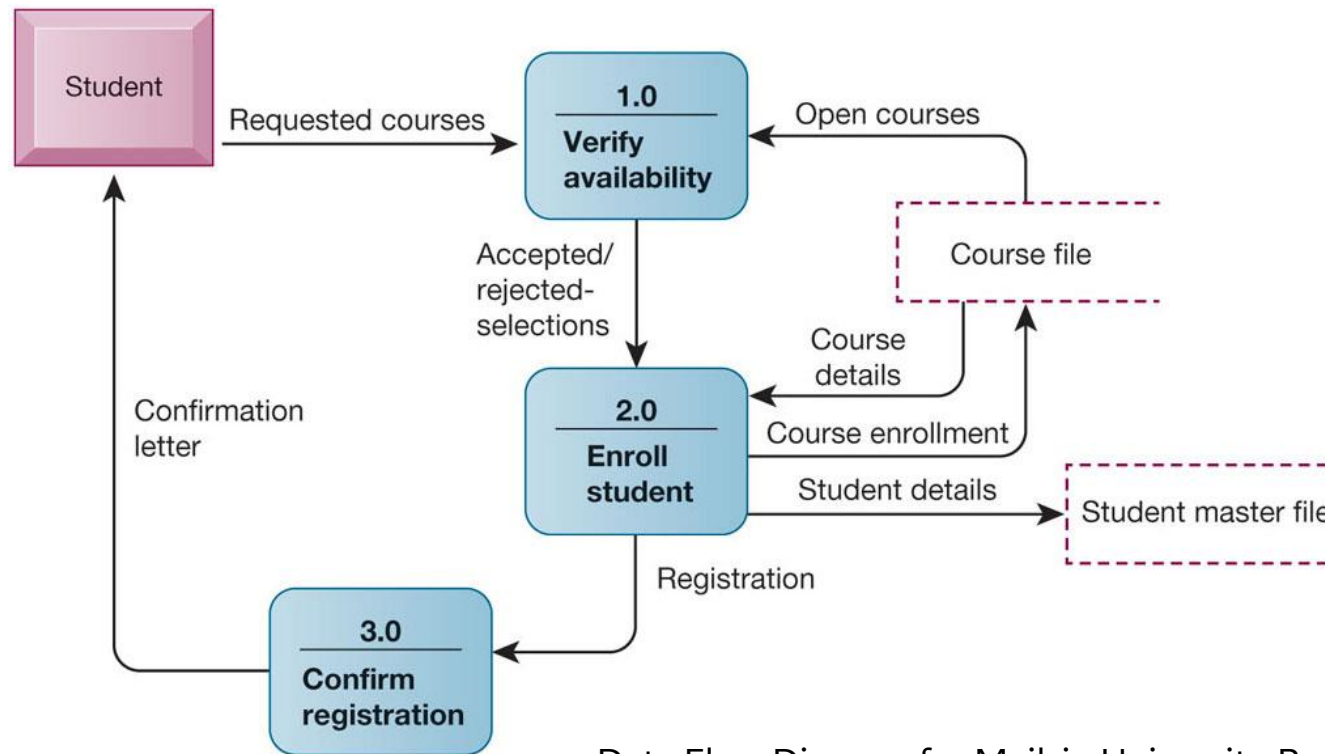
1. Structured methodologies
2. Object-oriented development

Structured methodologies

- Structured: Techniques are step-by-step, progressive
- Process-oriented: Focusing on modelling processes or actions that manipulate data
- Separate data from processes
- Data flow diagram (DFD)
 - Represents system's component processes and flow of data between them
 - Logical graphic model of information flow
- Data dictionary
 - Defines contents of data flows and data stores

Structured methodologies (cont.)

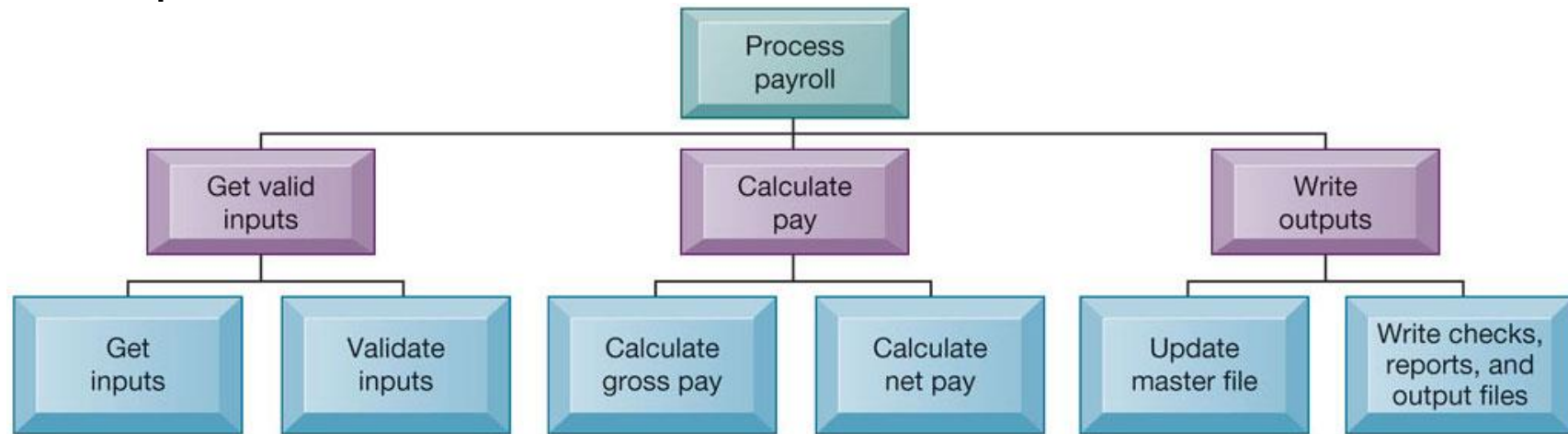
- Process specifications
 - Describe transformation occurring within lowest level of data flow diagrams



Data Flow Diagram for Mail-in University Registration System

Structured methodologies (cont.)

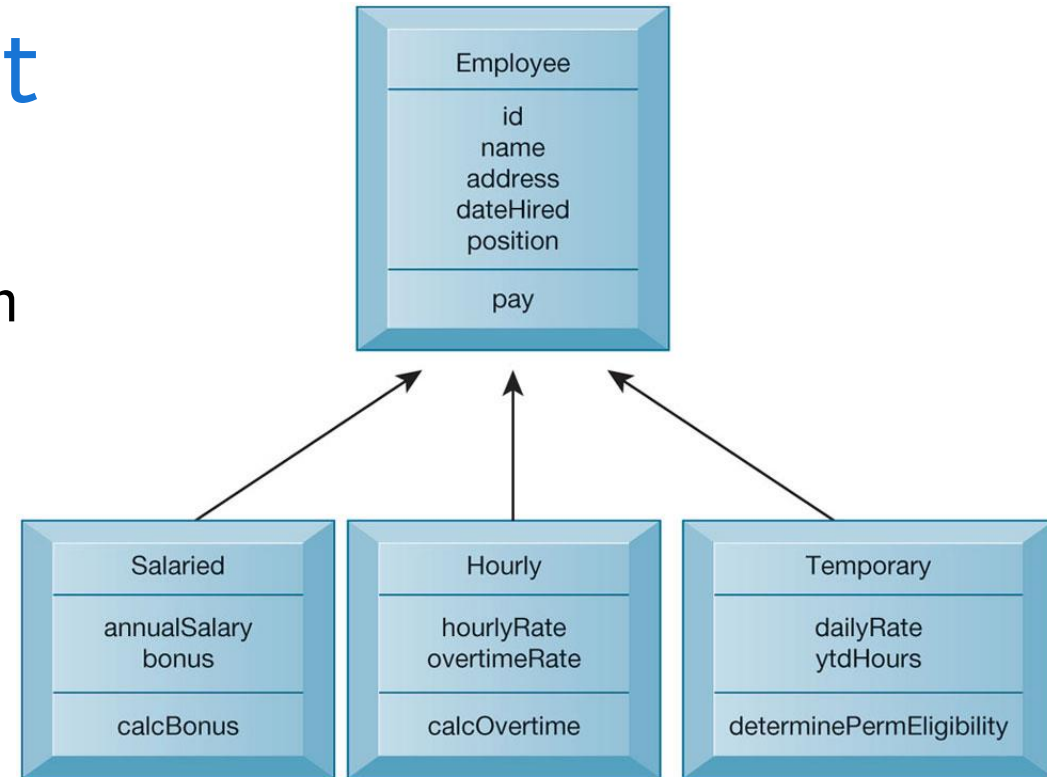
- Structure chart
 - Top-down chart, showing each level of design, relationship to other levels, and place in overall design structure
 - Example: diagram shows a high-level structure chart. This is high-level diagram and note that subfunctions exist for the functions “Get inputs,” “Validate inputs,” and so on.



High-level Structure Chart for a Payroll System

Object-Oriented Development

- Object
 - Basic unit of systems analysis and design
 - Combines data and the processes that operate on those data
 - Data in object can be accessed only by operations associated with that object
- Object-oriented modeling
 - Based on concepts of class and inheritance
 - Objects belong to a certain class and have features of that class
 - May inherit structures and behaviors of a more general, ancestor class

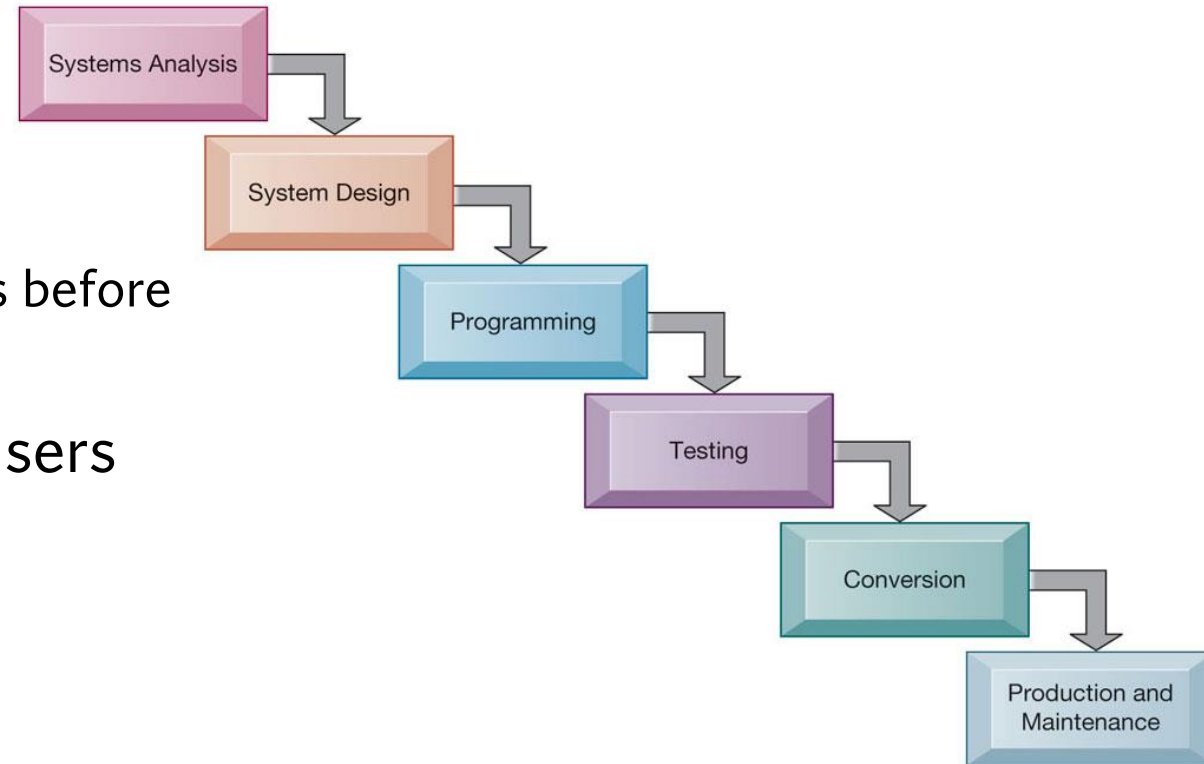


Object-Oriented Development (cont.)

- More iterative and incremental than traditional structured development
 - Systems analysis: Interactions between system and users analyzed to identify objects
 - Design phase: Describes how objects will behave and interact; grouped into classes, subclasses, and hierarchies
 - Implementation: Some classes may be reused from existing library of classes, others created or inherited
- Objects are reusable
 - Object-oriented development can potentially reduce time and cost of development

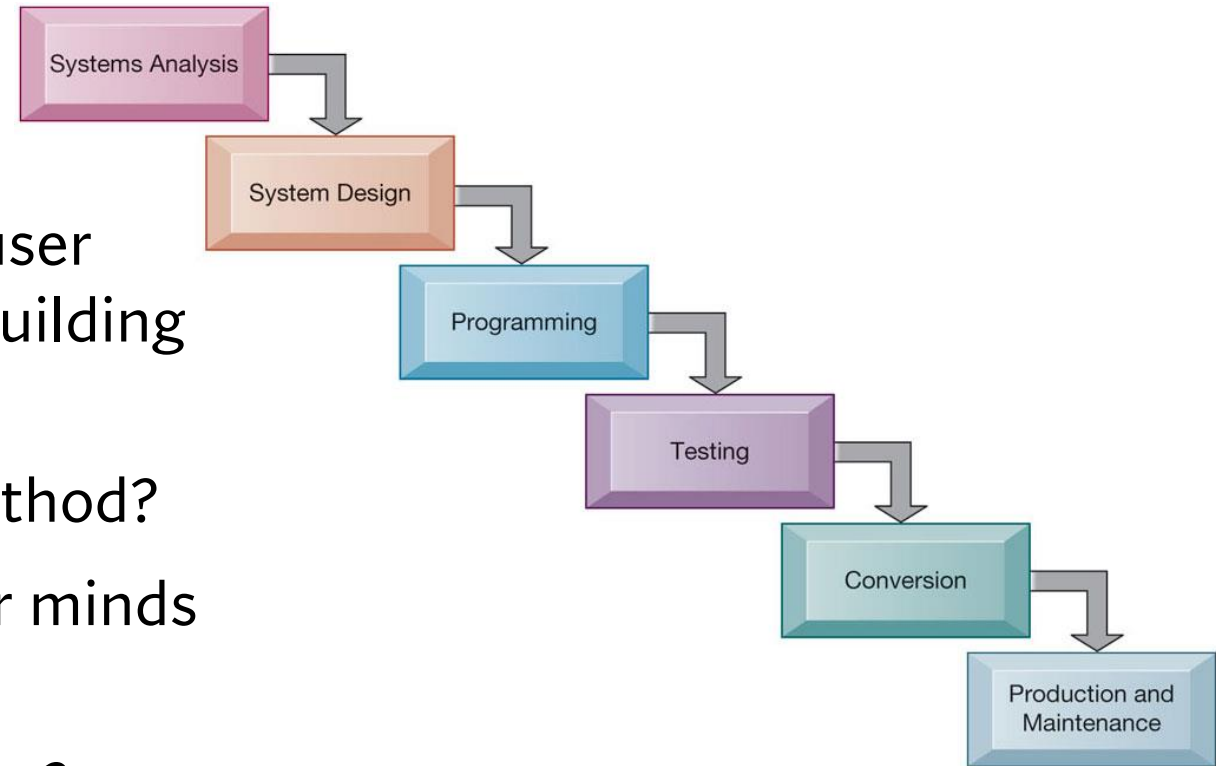
Traditional Systems Life Cycle

- Oldest method for building information systems
- Phased approach
 - Development divided into formal stages
 - “Waterfall” approach: One stage finishes before next stage begins
- Formal division of labor between end users and information systems specialists
- Emphasizes formal specifications and paperwork
- Still used for building large complex systems
- Can be costly, time-consuming, and inflexible



Discussion

- What are the effects of unanticipated user requirements when using this type of building method.
- What is the role of end users in this method?
- What happens when users change their minds halfway through the project?
- Can the systems analysis be started over?



Outsourcing

- Several types
 - Cloud and SaaS providers
 - Subscribing companies use software and computer hardware provided by vendors
- External vendors
 - Hired to design, create software
 - Domestic outsourcing
 - Driven by firm's need for additional skills, resources, assets
 - Offshore outsourcing
 - Driven by cost-savings
- Advantages
 - Allows organization flexibility in IT needs
- Disadvantages
 - Hidden costs, for example:
 - Identifying and selecting vendor
 - Transitioning to vendor
 - Opening up proprietary business processes to third party

Discussion

- The amount of work involved in partnering and sharing work with a vendor. It may take anywhere from three months to a year to fully transfer work to a vendor.
- What other types of hidden costs can students identify?

TOTAL COST OF OFFSHORE OUTSOURCING				
Cost of outsourcing contract		\$10,000,000		
Hidden Costs	Best Case	Additional Cost (\$)	Worst Case	Additional Cost (\$)
1. Vendor selection	.02%	20,000	2%	200,000
2. Transition costs	2%	200,000	3%	300,000
3. Layoffs & retention	3%	300,000	5%	500,000
4. Lost productivity/cultural issues	3%	300,000	27%	2,700,000
5. Improving development processes	1%	100,000	10%	1,000,000
6. Managing the contract	6%	600,000	10%	1,000,000
Total additional costs		1,520,000		5,700,000
	Outstanding Contract (\$)	Additional Cost (\$)	Total Cost (\$)	Additional Cost
Total cost of outsourcing (TCO) best case	10,000,000	1,520,000	11,520,000	15.2%
Total cost of outsourcing (TCO) worst case	10,000,000	5,700,000	15,700,000	57.0%

Example: Total Cost of Offshore Outsourcing

AI/ML in System Development

- Code Generation:
 - AI assistants like GitHub Copilot generate code from natural language
- Requirements Engineering:
 - Automated extraction and analysis of requirements
- Test Generation:
 - Creating test cases and data based on specifications
- Documentation:
 - Automatic generation of user and technical documentation
- UI/UX Design:
 - Generating interface designs based on requirements

AI/ML Applications in Business Systems

- **Customer Service:** AI chatbots handling complex inquiries
- **Content Creation:** Automated report generation and content writing
- **Data Analysis:** Generating insights and visualizations from raw data
- **Business Forecasting:** Predictive analytics for demand and trends
- **Process Mining:** Analyzing existing processes for automation opportunities
- **Decision Support:** Providing recommendations based on historical data

AI/ML Applications in Business Systems

- **Before**
 - 360,000 hours of lawyer time annually reviewing contracts
- **Solution**
 - AI system to extract data points and clauses from loan agreements
- **Technology**
 - Machine learning, NLP, and document understanding
- **Results**
 - Improved accuracy in contract review
 - Seconds instead of hours per document
 - Lawyers focus on exception handling and complex negotiations
 - Significant cost savings and faster processing



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Implementing Gen AI/ML in Organizations

Key Considerations

- Data quality and governance
- Integration with existing systems
- Workforce upskilling
- Ethical considerations
- Regulatory compliance

Implementation approach

- Start with high-value, low-risk processes
- Pilot projects with clear metrics
- Human-in-the-loop systems
- Progressive automation

The Future of System Development

- **Predictive maintenance:** Self-correcting systems; identify issues, diagnose problems, and implement corrections without human intervention
- **AI-augmented development:** Collaboration between humans and AI
- **Low-code/no-code proliferation:** Democratization of development
- **Intelligent automation:** Self-optimizing systems
- **Embedded ethics:** Responsible AI by design

**Thank you for your attention to the module TCX2005.
All the best for your Final Exam!**

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of Business Analytics together...!**